

Asymmetric Capital Buffer Control

Manuscript skeleton, 10 October 2026

1 Introduction

The question is how an asymmetric penalty on a capital shortfall shapes a bank's recapitalisation policy and the volatility of its capital buffer. The model treats recapitalisation as impulse control under a fixed issuance cost and a regulatory cap on risky exposure.

Table 1: Headlines.

ID	Headline	Model rows	Scope note
----	----------	------------	------------

2 Model

Table 2: Model results.

ID	Claim	Statement	Lean	Proof
----	-------	-----------	------	-------

3 Literature

Table 3: Papers read.

ID	Paper	What we rely on	Quote	Page	Rows
----	-------	-----------------	-------	------	------

4 Conclusions

Table 4 holds the conclusions that rest on model rows and on papers read in full. Table 5 holds every claim that a result is new and every other claim still waiting, and each of its rows names what it waits on.

Table 4: Conclusions.

ID	Statement	Rows
----	-----------	------

Table 5: Pending conclusions.

ID	Statement	Rows
----	-----------	------

A Proofs

Notation.

B Evidence and verification

Toolchain. Lean v4.32.0-rc1 with Mathlib at tag v4.32.0-rc1, commit 360da6fa66c1, pinned in lean/mathlib/lake-manifest.json. Every file under lean/mathlib/ is a build root. A theorem passes the axiom audit when it depends on no axiom outside propext, Classical.choice and Quot.sound.

Checks. verify.sh runs checks/verify_manuscript.py, checks/verify_mathlib.py and checks/verify_docs.py, and writes their output to VERIFICATION.md. lit/verify_lit.sh runs the suite of every literature record.

What Lean does not carry.

Illustrations. No row rests on SymPy or on numerical sampling.

C Terminology

Term	Symbol	Meaning
------	--------	---------

D Primitives and scope

E Refuted conjectures